Winston Industries LLC

Winston

Compiler Project – Spring 2026
Software Development Plan
Version <1.0>

[Note: The following template is provided for use with the Unified Process for EDUcation. Text enclosed in square brackets and displayed in blue italics (style=InfoBlue) is included to provide guidance to the author and should be deleted before publishing the document. A paragraph entered following this style will automatically be set to normal (style=Body Text).]

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Revision History

Date	Version	Description	Author
02-16-2026	0.1	Project Plan Draft	Davina Love

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Software Development Plan

1. Introduction

The purpose of this document is to outline the development of a small-scale compiler.

The compiler should be able to tokenize the user input and perform actions.

1.1 Purpose

The purpose of this plan is to provide clarity for the boundaries of the project and delegate roles as needed.

The following people use the *Software Development Plan*:

- The **project manager** uses it to plan the project schedule and resource needs, and to track progress against the schedule.
- **Project team members** use it to understand what they need to do, when they need to do it, and what other activities they are dependent upon.

1.2 Scope

This *Software Development Plan* describes the overall plan to be used by the Compiler project, including deployment of the product. The details of the individual iterations will be described in the Iteration Plans. The plans as outlined in this document are based upon the product requirements as defined in the *Vision Document*.

1.3 Definitions, Acronyms, and Abbreviations

See the Project Glossary.

1.4 References

1.5 Overview

This *Software Development Plan* contains the following information:

Project Overview — Compiler that tokenizes the user input and perform calculations

Project Organization — Team comprises of a team leader and team members, one of which is a meeting note-taker.

Management Process — The project should take the semester to complete.

Applicable Plans and Guidelines — Software Architecture Design, Software Requirements, User Manuals, Test-Cases

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

The compiler should be able to parse through user input and perform computations.

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2.2 Assumptions and Constraints

Project will be done in C++. Team is limited to their own personal devices / campus provided equipment. Programming environment is up to each team member's discretion.

2.3 Project Deliverables

Project Plan – 22 Feb 2026

Software Requirements Specifications – 15 March 2026

Software Architecture – 5 April 2026

Implementation, Test Case, User Manual – 7 May 2026

2.4 Evolution of the Software Development Plan

*[A table of proposed versions of the **Software Development Plan**, and the criteria for the unscheduled revision and reissue of this plan. The text below is provided as an example.]*

0.1	Tokenizer
0.2	Interpreter
0.3	Calculator
1.0	TBD

*Additional features will be added once the main components have been implemented.

3. Project Organization

3.1 Organizational Structure

Professor Hossein Saidien, GTA Toye Oloko, GTA Liangqin Ren

Project Leader

Assistant Project Leader

Quality Assurance

Technical Leader

Team Member

3.2 Roles and Responsibilities *[the more details here, the easier your job; include contact info, availability info, expertise, …]*

Person	**Unified Process for EDUcation Role**
Davina Love	Project Leader
Landrie Rolla	Assistant Project Leader
Christian Mitchell	Team Admin
Conner Magee	Quality Assurance
Liam Gunther	Technical Leader

Caden Gundenkauf	Team Member

Anyone on the project can perform Any Role activities.

4. Management Process

4.1 Project Plan

Project Plan – 22 Feb 2026

Software Requirements Specifications – 15 March 2026

Software Architecture – 5 April 2026

Implementation, Test Case, User Manual – 7 May 2026

4.1.1 Iteration Objectives

4.1.2 Releases

4.1.3 Project Schedule

- Project Plan
  - Setup Communications (4 Feb 2026)
  - Setup Project Organization (GitHub, Discord) (16 Feb 2026)
  - Initial Draft Made (16 February 2026)
- Software Requirements Specifications
  - Purpose, Scope, User Requirements (5 March 2026)
  - Use Case Specifications (15 March 2026)
- Software Architecture
  - Define Scope (30 March 2026)
  - Define Goals and Constraints in line with requirements document (30 March 2026)
  - Use Case and Performance (30 March 2026)
  - Module Diagrams (5 April 2026)

- Implementation
  - Tokenizer
  - Interpreter
  - Calculator
  - Define Edge Cases within Architecture (throughout project)
  - Total 1.0 Release (7 May 2026)
- Test Cases
  - Define Procedure for test cases (1 May 2026)
  - Expand on Edge cases developed during implementation (7 May 2026)
  - Separate into individual cases for testing (7 May 2026)
- User Manual
  - Introduction/Getting Started (27 April 2026)
  - Advanced Features and Troubleshooting (6 May 2026)

4.2 Project Monitoring and Control

- Requirements Management: Naming convention will be “ProjectName_ClassName_ModuleName”. A formal link to the product requirements will be provided with the included GitHub Repository. The Section

titled Software Requirements Specification will contain the product requirements themselves and will have an included change log formally outlining the changes made. A Discord Text Channel will be used for live updating on small updates. For measuring previous changes made to the requirements that may have not been detailed in error, the included version history in word (file>version history) will be used to compare versions and find changes.

- Quality Control: Discord channel will be used to post new versions along with processes to test functionality. A Link is provided to the Quality Control spreadsheet (made in excel) wherein a priority level is assigned to software fragments requiring implementation.
  Priority levels:
  High: Urgent bugs or Core software fragments that are included in or impact the products core features. Should be addressed within a few days.
  Medium: Small features or preventative maintenance needed for previous software fragments. Should be addressed within a week.
  Low: A comment or change to a comment. Not urgent or part of core features. Can be added immediately with little review.
  QC Procedures: Testing software integration in terminal, testing predicted affected feature outputs and functionality. Coordinates with test cases outlined in the test case documentation.
- Risk Management: All discovered bugs are listed and categorized based on risk chance caused by potential crashes and bugs, major and minor.
- Configuration Management: Spreadsheet on quality control channel that keeps track of bugs and functionality failures, organized by priority and level of impact on functionality. High priority being program breaking and low priority optional.
  - Backup: Project will be stored on each members device individually as well as stored on Microsoft cloud servers (OneDrive) and GitHub servers.

4.3 Quality Control

Defects will be recorded and tracked as Change Requests, and defect metrics will be gathered (see Reporting and Measurement below).

All deliverables are required to go through the appropriate review process, as described in the Development Case. The review is required to ensure that each deliverable is of acceptable quality, using guidelines and checklists.

Any defects found during review which are not corrected prior to releasing for integration must be captured as Change Requests so that they are not forgotten.

4.4 Risk Management

Risks will be identified in Inception Phase using the steps identified in the RUP for Small Projects activity “Identify and Assess Risks”. Project risk is evaluated at least once per iteration and documented in this table.

Refer to the Risk List Document (CCC-DDD-X.Y.doc) for detailed information.

4.5 Configuration Management

Appropriate tools will be selected which provide a database of Change Requests and a controlled versioned repository of project artifacts.

All source code, test scripts, and data files are included in baselines. Documentation related to the source code is also included in the baseline, such as design documentation. All customer deliverable artifacts are included in the final baseline of the iteration, including executables.

The Change Requests are reviewed and approved by one member of the project, the Change Control Manager role.

Refer to the Configuration Management Plan (EEE-FFF-X.Y.doc) for detailed information.

5. Annexes

The project will follow the UPEDU process.

Other applicable process plans are listed in the references section, including Programming Guidelines.